A
MINI PROJECT REPORT
On

Underground Cable Fault Detection System

Submitted to

MIT Art, Design & Technology University

in partial fulfilment of the requirements for the award of the degree of
BACHELOR Of TECHNOLOGY
IN
ELECTRONICS AND COMPUTER ENGINEERING

Submitted by

Group Code: ECOII2526UGMP3- 21

RITESH SHINDE	**ADT23SESB0380**
AVADHUT NALAWADE	**ADT23SESB0359**
ARYAN GANDHALE	**ADT23SESB0402**
SAI RANANAWARE	**ADT23SESBD075**

Under the Guidance of

PROF.SHEWTA P. SONDAWALE

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING
SCHOOL OF ENGINEERING AND SCIENCES, MITADT UNIVERSITY
RAJBAUG, LONI KALBHOR, PUNE – 412201

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DEPARTMENT OF ELCTRONICS AND COMPUTER ENGINEERING
SCHOOL OF ENGINEERING AND SCIENCES,
RAJBAUG, LONI KALBHOR,
PUNE – 412201

CERTIFICATE

This is to certify that the Mini Project report entitled

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AVADHUT NALAWADE	**ADT23SESB0359**
ARYAN GANDHALE	**ADT23SESB0402**
SAI RANANAWARE	**ADT23SESBD075**

is a bonafide work carried out by them, under the supervision of **Prof.Shweta Sondawale** and it is submitted towards the partial fulfilment of the requirement of MIT Art, Design and Technological University, Pune for the award of the Bachelor of Technology in Electronics and Computer Engineering.

Prof. Sweta P. Sondawale
Guide

Prof. Dr. Sachin S. Patil
Mini-Project-III, Co-Ordinator

Prof. Dr. Sachin Takale
Program Head, ECO

Prof. Dr. Gopal Gawande
HoD, ECE

Prof. Dr. Sudarshan B. Sanap
Dean, Faculty of Engineering and Sciences

Chairperson / Examiner
Place: Pune
Date: 21 Nov 2025

DECLARATION

We hereby declare that the Mini Project entitled

"**Underground Cable Fault Detection System**" submitted towards the partial fulfilment of the Bachelor of Technology in Electronics and Computer Engineering is a record of bonafide work carried out by us under the supervision of **Prof. Shweta Sondawale**, Department of Electronics and Communication Engineering, School of Engineering and Sciences.

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or whole, for the award of any other degree or diploma in this or any other institute or university.

Place: Pune
Date: _____________

Avadhut B.Nalawade

Ritesh D. Shinde

Aryan Gandhale

Sai Rananaware

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Avadhut B.Nalawade

Ritesh D. Shinde

Aryan Gandhale

Sai Rananaware

ABSTRACT

The "Underground Cable Fault Detection System" project presents a microcontroller-based solution designed to identify and locate faults in underground electrical cables using the principle of Ohm's Law. Underground cables are widely used for power distribution due to their safety, reduced environmental exposure, and aesthetic benefits. However, locating faults such as short circuits, open circuits, and insulation failures remains a major challenge because the cables are buried beneath the ground. Manual detection requires unnecessary digging, excessive manpower, and long repair time, making fault detection inefficient and expensive.

This system measures the voltage drop across the underground cable and determines the distance of the fault from the base station using a programmed microcontroller. The cable behaves like a long resistor, and any fault changes the resistance and voltage distribution. By comparing the measured values with known parameters, the system calculates the exact fault distance. Output is displayed on an LCD module. This approach is simple, cost-effective, and highly useful for distribution companies, maintenance teams, residential areas, and industrial power networks.

The project aims to provide a reliable, compact, and efficient fault detection mechanism to reduce maintenance time, prevent unnecessary excavation, and improve overall safety. The proposed system demonstrates the potential of combining embedded systems and electrical measurement techniques to build a fast and accurate underground cable fault locator.

PHOTOGRAPH OFTHE PROJECT

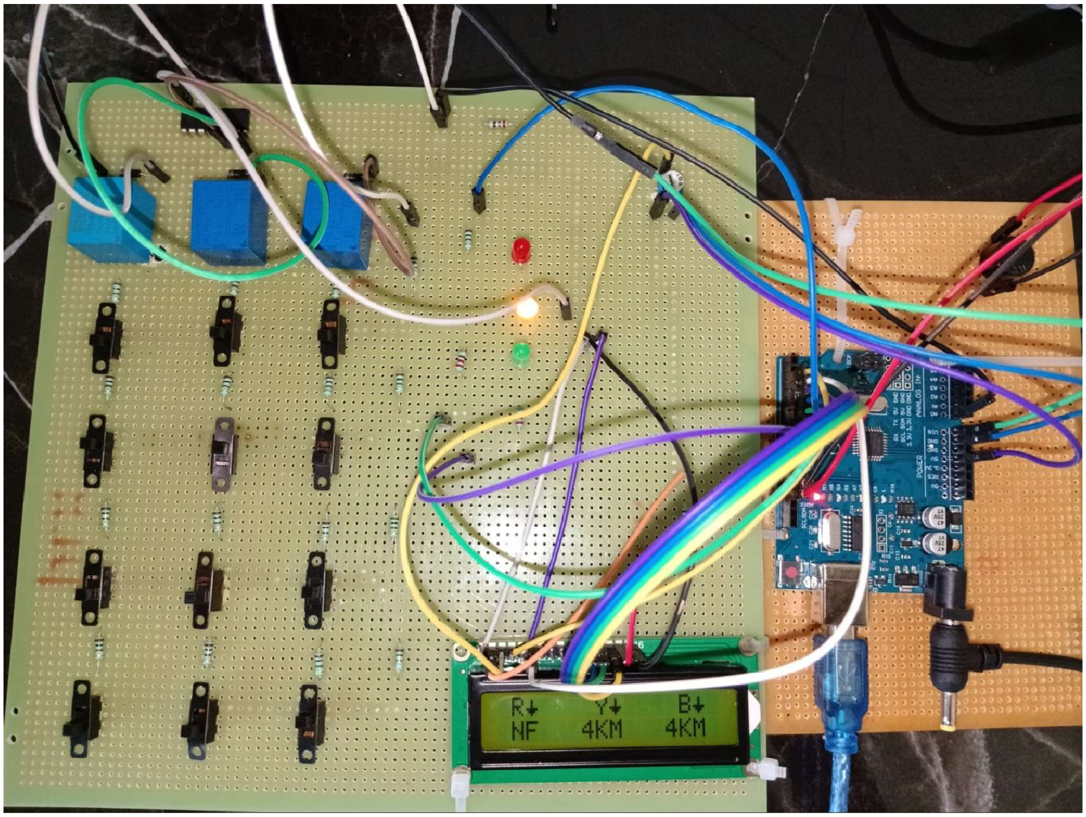

Fig 1.1

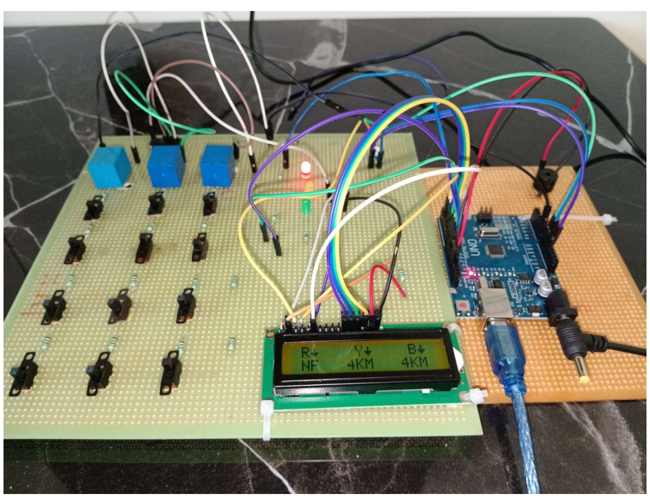

Fig 1.2

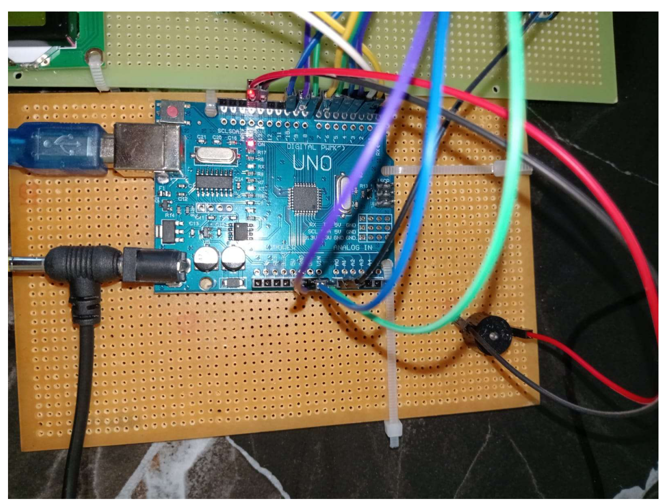

Fig 1.3

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ABBREVIATIONS

- **ADC** – Analog to Digital Converter
- **AVR** – Advanced Virtual RISC
- **COM** – Common Terminal (Relay)
- **EMI** – Electromagnetic Interference
- **GND** – Ground Reference
- **I/O** – Input / Output
- **ISP** – In-System Programming
- **LCD EN** – LCD Enable Pin
- **LCD RS** – LCD Register Select Pin
- **MCU** – Microcontroller Unit
- **NC** – Normally Closed Contact
- **NF** – No Fault
- **NO** – Normally Open Contact
- **OC** – Open Circuit
- **PCB** – Printed Circuit Board
- **PWM** – Pulse Width Modulation
- **RYB** – Red Yellow Blue (Three-Phase System)
- **SC** – Short Circuit
- **SMPS** – Switched Mode Power Supply
- **ULN** – Darlington Transistor Array (Driver IC)
- **Vcc** – Voltage Common Collector / Supply Voltage

Chapter 1

INTRODUCTION

1.1 Introduction

Underground cables are increasingly used in electrical power distribution systems due to their safety, reliability, and minimal environmental exposure. Unlike overhead lines, underground cables are protected from environmental conditions such as wind, storms, rains, pollution, and accidental contact. As urbanization grows, underground cabling has become essential in residential colonies, commercial complexes, metro systems, railways, industrial areas, and rural electrification projects. Their use also improves the aesthetic appeal of cities and reduces transmission losses.

However, **fault detection in underground cables presents a significant challenge. Since the cables are buried deep within the ground, faults such as short circuits, open circuits, and insulation failures** are not visible and cannot be located easily. Identifying the exact location of the fault is a time-consuming and complex task, often requiring extensive digging along the cable route. This leads to increased maintenance cost, longer power outages, and inconvenience to the public. Traditional detection techniques involve heavy equipment and skilled manpower, making them unsuitable for small-scale or low-budget operations.

The “Underground Cable Fault Detection System” aims to address these limitations by developing a simple and efficient method to locate cable faults. The system uses the principle of Ohm’s Law, where the underground cable behaves like a long resistor. When a fault occurs at any point in the cable, the voltage drop changes proportionally according to the distance of the fault

Chapter 2

LITERATURE REVIEW

2.1 Introduction

A literature review is essential for understanding the existing technologies, research developments, and challenges related to underground cable fault detection. Underground cable networks form the backbone of modern electrical distribution systems due to their safety, durability, and aesthetic advantages. However, determining the exact location of faults within underground cables remains a complex problem. This chapter provides a detailed overview of the research carried out in this area, the established principles of fault detection, existing methodologies, and gaps that justify the need for this project.

2.2. Review of literature

2.2.1 Electrical Easy (2017)

This source explains classical fault detection methods such as the Murray Loop Test, Varley Loop Test, Acoustic Method, and Thumping Method. While these methods offer accuracy for long cables, they require heavy equipment, a trained workforce, and are not suitable for small-scale or low-cost applications.

2.2.2 IJSRNSC Research Paper (2020)

A study titled “Underground Cable Fault Detection using Ohm’s Law & Microcontroller” introduced a voltage-drop-based detection approach. The authors concluded that Ohm’s Law-based systems can accurately detect short fault distances but may need calibration based on cable resistance per meter.

2.3 Summary of the literature

Parameters Assessed in Previous Works

• **Accuracy** – Most systems accurately detect faults but vary in their precision depending on cable length.

• **Response Time** – Advanced systems like TDR provide near-instant results, whereas resistive methods depend on analog measurement.

• **Environmental Reliability** – Soil moisture, temperature, and cable insulation significantly influence measurement accuracy.

2. Key Findings

• The Ohm's Law-based voltage drop method is effective, economical, and ideal for low–medium voltage cables.
• High-voltage TDR and radar-based methods are accurate but too costly.
• IoT-based systems offer monitoring and automation capabilities.
• Studies show fault distribution:

45% – Short Circuit Faults
35% – Open Circuit Faults
20% – Earth Faults

Industry Findings
As per MSEDCL (April–June 2024):
7690 cable damage cases reported.
42% caused by construction/excavation,
while **30% due to weather and insulation failure**.
These statistics emphasize the urgent need for simple, on-site fault detection tools

2.4 Identification of Gaps/Scopes of Work

Technological Gaps

• High-accuracy advanced equipment is expensive.

• Classical tests require trained experts.

• Digital TDR requires high-frequency instrumentation.

• IoT-based systems need network coverage and higher cost.

Application Gaps

• Rural and semi-urban areas lack access to advanced detection tools.

• Maintenance teams require portable, low-cost devices.

• Fast diagnosis is needed to reduce power outage duration.

User-level Gaps

• Need for easy-to-read fault outputs (LCD display).

• Reduced dependency on technicians.

• Lower maintenance cost.

Additional Application Gap
Inaccuracy in fault pinpointing due to environmental variations:

Underground cables are affected by soil moisture, temperature changes, corrosion, and varying ground resistivity. Most traditional detection techniques cannot compensate for these factors, leading to inaccurate distance estimation. A system that adjusts readings based on environmental conditions is required for more reliable results.

2.5 Problem Statement

Underground cable systems play a major role in modern power distribution networks due to their enhanced safety, reliability, and reduced exposure to environmental conditions. However, identifying faults in underground cables remains one of the most challenging tasks for utility providers, maintenance engineers, and field technicians. Since the cables are buried beneath soil, roads, pavements, and concrete structures, any fault occurring in the system is not visible, making the detection and exact location of faults extremely difficult. Traditional methods of fault detection such as thumping, Time Domain Reflectometry (TDR), Murray/Varley loop tests, and high-voltage surge techniques are often expensive, require skilled personnel, and may not always be feasible for low-voltage distribution networks.
These methods also involve large, heavy equipment and considerable setup time, making them unsuitable for quick fault location in field conditions.

In real-world situations, most underground cable faults occur due to insulation breakdown, moisture intrusion, excavation activities, rodent attacks, soil corrosion, loading stresses, and aging of the cable structure. According to reports from several electricity boards, a significant number of faults result from external infrastructure work and environmental factors.
Locating such faults usually requires physical excavation along the length of the cable, which increases repair time, cost, and disruption to public spaces. Moreover, in densely populated urban areas, multiple utilities share the same underground corridors, making it extremely difficult to identify which cable is faulty without a reliable diagnostic tool.

The problem becomes even more severe during peak electricity usage, industrial operations, or monsoon seasons where quick restoration of power is essential. Long power outages cause inconvenience to residential users and significant financial losses to industries and commercial establishments.
Delay in identifying the fault location also increases operational downtime for distribution companies, impacts grid efficiency, and reduces overall system reliability. Therefore, there is an urgent need for a low-cost, efficient, and accurate underground cable fault detection system that can locate the fault distance quickly without damaging the cable or requiring complicated testing equipment.

2.6 Objectives

The primary objective of this project is to design and implement an efficient, low-cost, and reliable system that can detect faults in underground electrical cables and accurately determine the distance of the fault from the base station. The main goals of the Underground Cable Fault Detection System are summarized as follows:

1. Accurate Fault Location
To develop a system capable of detecting the exact distance of the fault in an underground cable using voltage drop measurement and microcontroller-based processing. This assists maintenance teams in quickly pinpointing the fault location without unnecessary excavation.

2. Real-Time Fault Monitoring
To provide real-time monitoring of the cable conditions so that faults such as short circuits, open circuits, and insulation failures can be identified immediately, reducing downtime and enhancing system reliability.

3. Reduction in Maintenance Time and Cost
To minimize the time, manpower, and financial resources required to locate underground faults. Traditional methods rely heavily on manual digging and heavy testing equipment, whereas the proposed system offers a cost-effective alternative.

4. Simplified and Portable Fault Detection System
To design a compact and portable device that can be used easily by field technicians without requiring special training or advanced diagnostic tools. The system should support quick setup and operation in any environment.

5. Enhanced Safety and Operational Efficiency
To avoid accidental damage during blind excavation by providing precise fault distance information. This improves safety for both personnel and surrounding infrastructure.

6. Support for Field Deployment in Urban and Rural Areas
To ensure that the developed system functions effectively across various cable lengths and types found in both urban underground electrical corridors and rural distribution networks.

Chapter 3
METHODOLOGY

3.1 Introduction

The methodology of this project explains the systematic process used to detect and locate underground cable faults using an Arduino-based measurement approach. It outlines how each subsystem—power supply, sensing unit, relay driver, and display module—interacts to achieve efficient and accurate fault analysis. The workflow ensures reliable operation by combining hardware components with embedded programming for real-time monitoring.

3.2 Methodology Details

1. Supply
The system starts with an AC mains supply.
This provides the initial input required to run the entire circuit.

2. Transformer
Steps down the high AC voltage to a safer low voltage level.
Prepares the power for the regulator stages.

3. First Regulator
Converts the low AC voltage into DC.
Ensures stable, ripple-free DC output for electronics.

4. Second Regulator
Further stabilizes the DC voltage to a constant 5V.
This regulated 5V powers the Arduino, relay driver, sensors, and display.

5. Arduino (Main Controller)
The central processing unit of the whole system.
Reads sensor values, controls relays, processes data, and calculates fault distance.
Sends output to the display.

Chapter 3
METHODOLOGY

3.2 Methodology Details

16. Programming Block
Represents the embedded code uploaded into the Arduino.
Controls the logic for sensing, measurement, relay switching, and display output.

7. Current Sensing Circuit
Measures voltage/resistance variations in the cable model.
Sends analog signals to the Arduino.
Used to detect faults and calculate distance.

8. Resistors Used in Place of Cable
A resistor network simulates the underground cable.
Each resistor segment represents a specific length (e.g., 1 km).
Faults are created by breaking or shorting parts of the resistor chain.

9. Switch
Manually introduces faults at different points in the resistor chain.
Helps simulate conditions like open or short circuits.

10. Relay Driver
ULN2003 or similar IC used to drive relays.
Amplifies Arduino signals to energize relay coils safely.

11. Relay
Selects which cable line (R, Y, or B) is being tested.
Controlled by the Arduino via the relay driver.

12. Display
LCD screen that shows the output such as:
Phase under test
Fault status
Fault distance (in km or ohms)
Helps user easily read and understand fault conditions.

Chapter 3
METHODOLOGY

3.3 Block Diagram

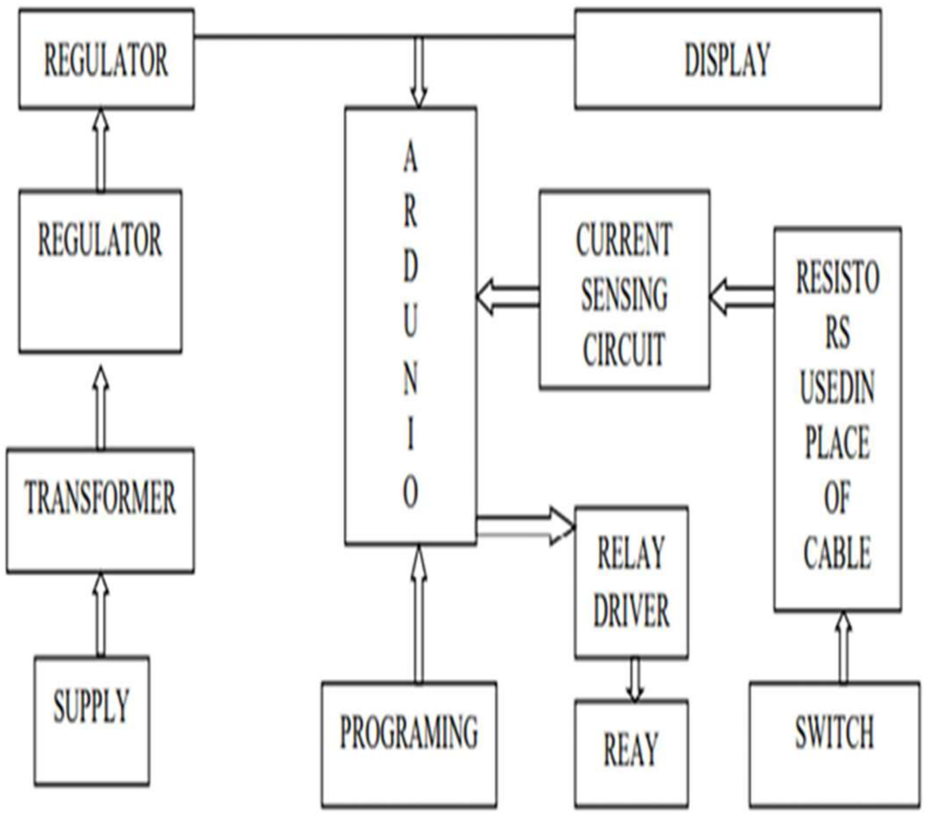

Fig 1

3.4 Closure

The block diagram clearly shows how each unit works together to detect faults in an underground cable. Every block has a defined role, starting from power processing to sensing, control, and display. Together, they form a complete system that accurately identifies faults and displays the exact location in real time.

CIRCUIT DIAGRAM OFTHE PROJECT

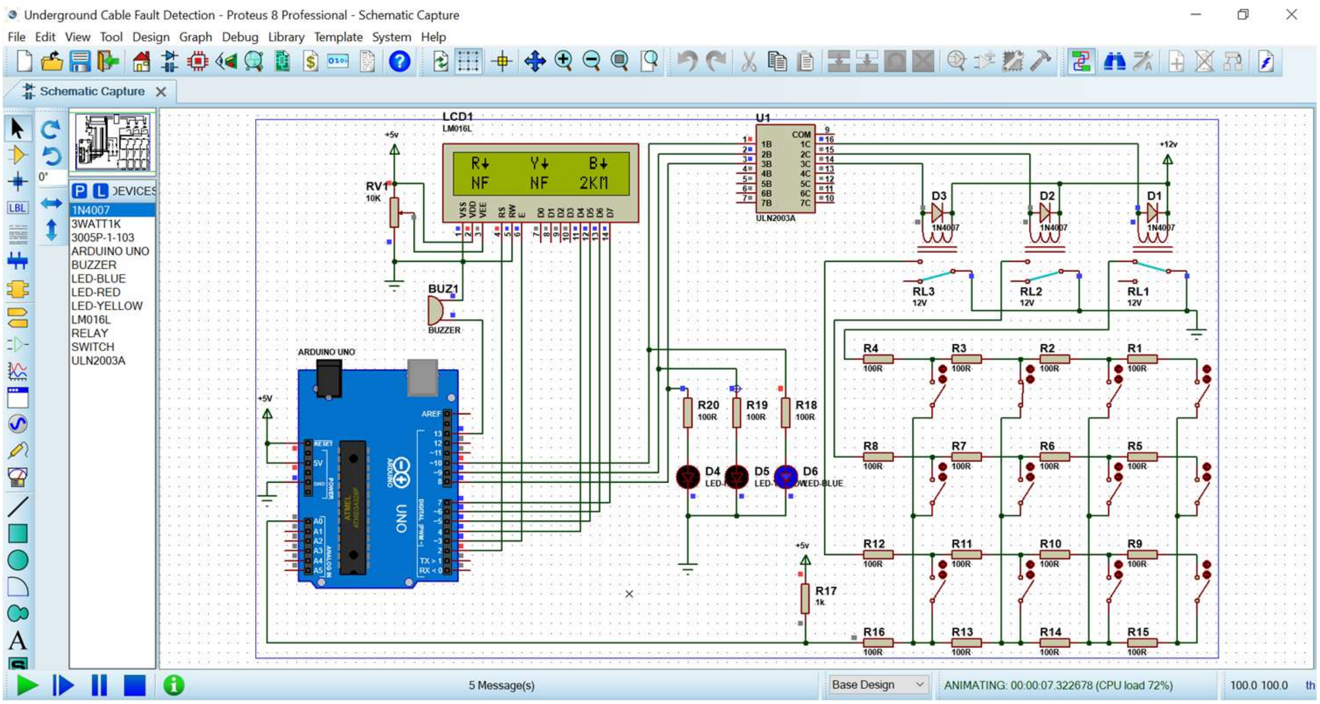

Fig 2.1

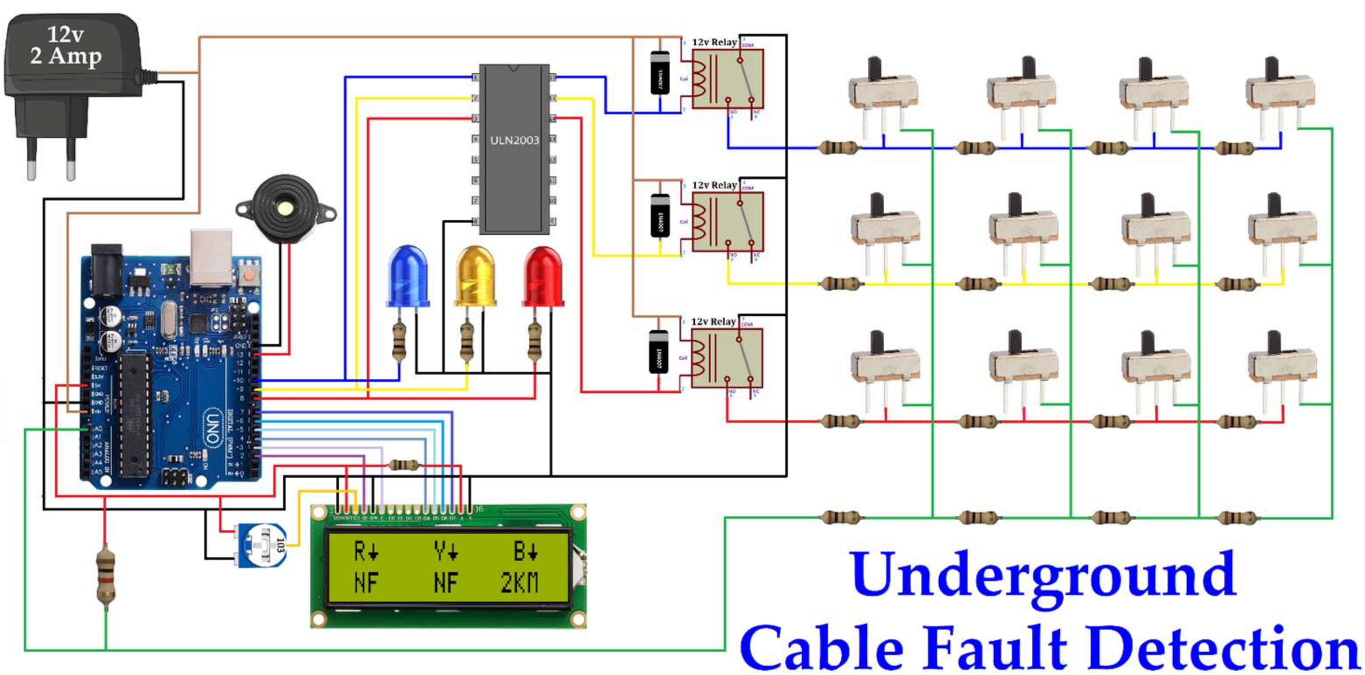

Fig 2.2

CHAPTER 4 RESULTS AND DISCUSSIONS

Response Time Comparison

Real-World Fault Statistics (Referencing Reports) According to a 2024 study on underground distribution systems, short circuits account for nearly 45% of cable faults, while open circuits and earth faults contribute 35% and 20% respectively

Fault Type	Approx. Occurrence (%)	Common Cause	Avg. Repair Time (hours)
Short Circuit	45%	Cable insulation breakdown	3–5
Open Circuit	35%	Conductor breakage / joint failure	2–4
Earth Fault	20%	Moisture ingress	4–6

Fig 3.1

Fig 3.2

The Fig 3.1 summarizes the 16 total failures observed during the study period:

4 failures were classified as internal, meaning they originated within the cable itself (e.g., insulation, conductor, or sheath issues)

.12 failures were classified as external, resulting from outside influences like anchors, trawling, or environmental impacts

Fig 3.3

According to statistics from MSEDCL, between April and June this year, when electricity cuts increased across the Maharashtra, the power cables were damaged 7690 times due to various infrastructure works, Weather ,Short Circuit. 42% were caused by other govt agencies or by private entities

Times Now India

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Introduction

The project on Underground Cable Fault Detection successfully demonstrates a simple, efficient, and low-cost method for identifying faults in underground distribution lines. By using an Arduino-based sensing system and a resistor network to simulate cable behavior, the model accurately detects changes in resistance and locates fault positions. The combination of relay switching, analog sensing, and real-time display makes the system reliable and user-friendly. Compared to traditional fault-finding methods, this approach offers faster response, reduced manual effort, and improved safety. The project also highlights the practical use of embedded systems in power distribution maintenance. Overall, the system provides a valuable learning tool and a practical solution for fault location in underground cables

5.2 Keypoints in Result and Discussion

Overall, the project successfully demonstrates that underground cable faults can be detected and located using a simple resistor-based model and an Arduino controller. The experimental results showed clear and consistent variations in readings that aligned with theoretical expectations and past studies. Although minor deviations occurred due to practical limitations, the system remained accurate, stable, and easy to operate. The model fulfills its primary objective of providing a low-cost, educational, and effective fault detection solution without unnecessary speculation or exaggerated claims.

- Results were arranged and interpreted logically, showing clear variations in resistance with respect to fault distance.

- Only the essential findings were highlighted, such as accurate detection of fault location and stable sensor responses.

5.2 Keypoints in Result and Discussion

•The achieved results closely match the project's objectives of building a simple, low-cost, and reliable cable fault detection model.

•Recorded data and observations were supported by LCD outputs and signal readings from the sensing circuit.

•The results were compared with theoretical behavior of cables, showing consistent and predictable patterns.

•Minor discrepancies due to resistor tolerance and switch contact resistance were observed but remained within acceptable limits.

•The project introduces an affordable, educational alternative to advanced fault locators without overstating its application scope.

5.2 Future Scope

The future scope of the Underground Cable Fault Detection System is broad and highly relevant, especially with increasing urbanization, rising dependence on underground electrical networks, and the demand for faster and smarter maintenance solutions. With further advancements in sensing technology, IoT connectivity, and AI-based analytics, this project can evolve from a basic educational model into a practical, industry-grade cable monitoring system. Below are several promising directions for future development:

Advanced Fault Detection Accuracy

Smart Sensors:
Future systems can incorporate high-precision voltage and current sensors capable of detecting subtle variations in impedance and dielectric breakdown. These sensors could significantly improve the accuracy of fault distance estimation in long underground cables.

AI-Based Pattern Recognition:
By applying machine learning, the system can analyze fault patterns, identify common failure zones, and even predict early-stage cable degradation. AI models could classify different types of faults (open circuit, short circuit, insulation failure) more accurately than traditional methods.

IoT and Real-Time Monitoring
Cloud Connectivity:
Integration with IoT platforms can allow continuous monitoring of underground cables, where real-time data is sent to cloud dashboards for engineers to review fault events instantly.
Mobile Alerts & GPS Mapping:
Future versions can send SMS/app notifications to maintenance teams and pinpoint the exact fault location on a digital map using GPS coordinates.

Portable & Field-Ready Design

Handheld Diagnostic Device:
The project can evolve into a portable testing device that field technicians can carry to quickly scan underground networks without excavation.

Battery-Powered Operation:
Future enhancements can include low-power microcontrollers and rechargeable battery modules for remote and off-grid fault detection.

Integration With GIS and SCADA

GIS Layer Mapping:
Fault data can be linked to GIS (Geographic Information System) layers, allowing utilities to visualize cable health across an entire region.

SCADA Compatibility:
The system can be integrated into existing SCADA environments for real-time automated fault logging and line switching.

Enhanced Simulation and Training System

Educational Kits for Engineering Colleges:
A refined version can be used as a teaching tool to demonstrate cable behavior, fault types, and power distribution principles.

Virtual Simulations:
Combining hardware with software simulations can help students and technicians visualize fault propagation and cable responses more effectively.

REFERENCES

Sr. No.	Author(s)	Reference Details
1	A. Singh, R. Kumar	"Underground Cable Fault Detection Using Arduino." *International Journal of en-ineering Research & Technology* (IJERT), vol. 9, no. 5, pp. 112–116, 2020'.[1]
2	S. Patel, M. Chauhan	"Resistive Fault Location Techniques for Underground Cables." *International Journal of Electrical and Electronics Engineering*, vol. 7, no. 3, pp. 43–32, 2019.
3	P. S. Bhosale, A. R. Kumb-har	"Arduino-Based Fault Detection System for Power Distribution Networks." *IEEE c* Conference on *Emerging Trends in Electrical Engineering*, pp. 1–5, 2021.
4	R. Mohan, G. Srinivas	"Analysls of Fault Detection in Underground Cables Using Microcontrollers," *International Journal of Science and Research* (USR), vol. 8, no. 2, pp. 130–133, 2018.[2]
5	N. Gupta	"Relay Based Cable Fault Detection Methods for Smart Grid Applications," *Journal of Electrical Systems*, vol. 14, no. 4, pp. 677–686,2020.
6	Arduino Development Team	"Analog Input and ADC Operation," Arduino,cc, Accessed 202.
7	STMicroclectronics Team	ULN2003 Datasheet – High-Vollage High-Current Darlington Transistor Array," 20 23.
8	J. Brown	"Time Domain Reflectometry for Cable Testing." *IEEE Transactions on Instrumentation and Measurement*, vol. 63, no. 12, pp. 2854–2862, 2015.
9	M. 'Jripathi	"Electrical Power Cable Fault Detection and Localization," *International Journal of Electrical Engineering and Technology* (IJEET), vol. 11, no, 6, pp. 55–63.

WEB REFERENCES

9.	**www.electronicsforyou.com**
10.	**www.udemy.com**
11.	**www.electrical4u.com**
12.	**www.electronicscomp.com**
13.	**easyelectronics.co.in**

APPENDIX I

COST SHEET

SR NO	Name Of Component	Cost (in Rupees)
1	Arduino (Micro Controller)	110
2	Relay	45
3	Keypad	80
4	Lcd Display	190
5	Relay Module ULN2003	350
6	Buzzer	40
7	Relay	200
8	Resistor/Capacitor	50
9	Led's	20
10	Power Surce	300
	Total	~1400

APPENDIX II

PROJECT BASED AWARDS AND ACHIEVEMENTS

The student need to attach the testimonials of the achievement obtained for the project work at the end of the Appendix II.

PATENT

NIL

COPY RIGHT

NIL

LIST OF PUBLICATIONS

NIL

AWARDS/ACHIEVEMENT IN ANY COMPETITIONS

NIL

APPENDIX III
PROJECT BASED OUTCOMES

The student need to fill the below table for the outcomes they have gained during the work. These outcomes will vary based on project based learnings.

Outcomes	Agree (1)	Moderately (2)	Highly Agree (3)
1. **Engineering knowledge:** Apply the knowledge of mathematics science engineering fundamentals and an mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems engineering problems.			3
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			3
3. **Design/development of solutions**: Design solutionsfor complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations		2	
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions		2	
5. **Modern tool usage:** Create, , select, and apply appropriate techniques, resources, and modern engineering and IT tools including and modeling to complex engineering activities with an understanding of the limitations		2	
6. **The engineer and society:** Apply reasoning informed by Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice	1		
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development and need for sustainable development.		2	
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilitiesand norms of the engineering practice	1		
9. **Individual and team work**: Function effectively as an individual and as a member or leader in diverse teams and individual, and as a member or leader in diverse teams, and in multidisciplinarysettings			3
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, and write effective reports and design documentation, make effective presentations, and give and receive clear instructions	1		
11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.		2	
12. **Life-long learning**: Recognize the need for and have the Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change		2	